LETRAM OS
LEVETIRACETAM ORAL SOLUTION 100MG/ML

Description
LETRAM OS (Levetiracetam Oral Solution 100mg/mL) 150mL Pack Size: A Clear, colourless grape flavoured liquid filled in 200mL amber coloured glass bottle.

LETRAM OS (Levetiracetam Oral Solution 100mg/mL) 300mL Pack Size: A clear, colorless grape flavoured liquid filled in 300ml amber coloured glass bottle with EPE wad.

Each ml contains 100.00mg of Levetiracetam, 1.500mg of Methyl Parahydroxy Benzoate (as preservative) and 0.180mg of Propyl Parahydroxy Benzoate (as preservative).

Pharmacodynamics

The active substance, Levetiracetam is a pyrrolidone derivative (S-enantiomer of α-ethyl-2-oxo-1-pyrrolidine acetamide), chemically unrelated to existing antiepileptic active substances.

Mechanism of action

The mechanism of action of levetiracetam still remains to be fully elucidated but appears to be different from the mechanisms of current antiepileptic medicinal products.

Pharmacodynamics effects

Levetiracetam induces seizure protection in a broad range of animal models of partial and primary generalized seizures without having pro-convulsant effect. The primary metabolite is inactive.

In man, activity in both partial and generalized epilepsy conditions (epileptiform discharge/photo paroxysmal response) has confirmed the broad spectrum of the preclinical pharmacological profile.

Pharmacokinetics

Levetiracetam is a highly soluble and permeable compound. The pharmacokinetic profile is linear and time-independent with low intra- and inter-subject variability. There is no modification of the clearance after repeated administration. There is no evidence for any relevant gender, race or circadian variability. The pharmacokinetic profile is comparable in healthy volunteers and in patients with epilepsy.

Due to its complete and linear absorption, plasma levels can be predicted from the oral dose of levetiracetam expressed as mg/kg bodyweight. Therefore there is no need for plasma level monitoring of levetiracetam.

A significant correlation between saliva and plasma concentrations has been shown in adults and children (ratio of saliva/plasma concentrations ranged from 1 to 1.7 for oral tablet formulation and after 4 hours post-dose for oral solution formulation).

Adults and adolescent

Absorption

Levetiracetam is rapidly absorbed after oral administration. Oral absolute bioavailability is close to 100%. Peak plasma concentrations (C_{max}) are achieved at 1.3 hours after dosing. Steady-state is achieved after two days of a twice daily administration schedule. Peak concentrations (C_{max}) are typically 31 and 43 µg/ml following a single 1,000 mg dose and repeated 1,000 mg b.i.d. dose, respectively. The extent of absorption is dose-independent and is not altered by food.

Distribution

No tissue distribution data are available in humans. Neither levetiracetam nor its primary metabolite are significantly bound to plasma proteins (< 10%). The volume of distribution of levetiracetam is approximately 0.5 to 0.7 l/kg, a value close to the total body water volume.

Biotransformation

Levetiracetam is not extensively metabolized in humans. The major metabolic pathway (24% of the dose) is an enzymatic hydrolysis of the acetamide group. Production of the primary metabolite, ucb L057, is not supported by liver cytochrome P_{450} isoforms. Hydrolysis of the acetamide group was measurable in a large number of tissues including blood cells. The metabolite ucb L057 is pharmacologically inactive.

Two minor metabolites were also identified. One was obtained by hydroxylation of the pyrrolidone ring (1.6% of the dose) and the other one by opening of the pyrrolidone ring (0.9% of the dose).

Other unidentified components accounted only for 0.6% of the dose.

No enantiomeric interconversion was evidenced *in vivo* for neither levetiracetam nor its primary metabolite.

In vitro, levetiracetam and its primary metabolite have been shown not to inhibit the major human liver cytochrome P450 isoforms (CYP3A4, 2A6, 2C9, 2C19, 2D6, 2E1 and 1A2), glucuronyl transferase (UGTIA1 and UGTIA6) and epoxide hydroxylase activities in addition, levetiracetam does not affect the *in vitro* glucuronidation of valproic acid.

In human hepatocytes in culture, levetiracetam had little or no effect on ethinylestradiol conjugation or CYP1A1/2. Levetiracetam caused mild induction of CYP2B6 and CYP3A4 at high concentrations (680 µg/ml), however at concentrations approximating to the C_{max} following a repeated 1500mg twice daily dose, the effects were not considered to be biologically relevant. Therefore, the interaction of LEVETIRACETAM with other substances, or vice versa, is unlikely.

Elimination

The plasma half-life in adults was 7±1 hours and did not vary either with dose, route of administration or repeated administration. The mean total body clearance was 0.96 ml/min/kg.

The major route of excretion was via urine, accounting for a mean 95% of the dose was excreted within 48 hours). Excretion via faeces accounted for only 0.3% of the dose.

The cumulative urinary excretion of levetiracetam and its primary metabolite accounted for 66% and 24% of the close, respectively during the first 48 hours. The renal clearance of levetiracetam and ucb L057 is 0.6 and 4.2 ml/min/kg respectively indicating that levetiracetam is excreted by glomerular filtration with subsequent tubular reabsorption and that the primary metabolite is also excreted by active tubular secretion in addition to glomerular filtration. Levetiracetam elimination is correlated to creatinine clearance.

Elderly

In the elderly, the half-life is increased by about 40% (10 to 11 hours). This is related to the decrease in renal function in this population.

Children (4 to 12 years)

Following single dose administration (20 mg/kg) to epileptic children, the half-life of levetiracetam was 6.0 hours. The apparent body clearance was 1.43 ml/min/kg. Following repeated oral dose administration (20 to 60 mg/kg/day) to epileptic children (4 to 12 years), levetiracetam was rapidly absorbed. Peak plasma concentration was observed 0.5 to 1.0 hour after dosing. Linear and dose proportional increases were observed for peak plasma concentrations and area under the curve. The elimination half-life was approximately 5 hours. The apparent body clearance was 1.1 ml/min/kg.

Infants and children (1 month to 4 years)

Following single dose administration (20 mg/kg) of a 100 mg/ml oral solution to epileptic children (1 month to 4 years), levetiracetam was rapidly absorbed and peak plasma concentrations were observed approximately 1 hour after dosing. The pharmacokinetic results indicated that half-life was shorter (5.3 h) than for adults (7.2 h) and apparent clearance was faster (1.5 ml/min/kg) than for adults (0.96 ml/min/kg).

Renal impairment

The apparent body clearance of both levetiracetam and of its primary metabolite is correlated to the creatinine clearance. It is therefore recommended to adjust the maintenance daily dose of LEVETIRACETAM based on creatinine clearance in patients with moderate and severe renal impairment.

In anuric end-stage renal disease subjects the half-life was approximately 25 and 3.1 hours during interdialytic and intradialytic periods, respectively.

The fractional removal of levetiracetam was 51% during a typical 4-hour dialysis session.

Hepatic impairment

In subjects with mild (Child-Pugh B) hepatic impairment, the pharmacokinetics of levetiracetam were unchanged. In subjects with severe hepatic impairment (Child-Pugh C), total body clearance was 50% that of normal subjects, but decreased renal clearance accounted for most of the decrease.

No dose adjustment is needed in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the renal insufficiency. Therefore a 50% reduction of the daily maintenance dose is recommended when the creatinine clearance is $<60 ml/min/1.73m^2$

Indications

LEVETIRACETAM is indicated as monotherapy in the treatment of partial onset seizures with or without secondary generalization in patients from 16 years of age with newly diagnosed epilepsy.

LEVETIRACETAM is indicated as adjunctive therapy:

- In the treatment of partial onset seizures with or without secondary generalization in adults and children from 4 years of age with epilepsy.
- In the treatment of myoclonic seizures in adults and adolescents from 12 years of age with Juvenil Myoclonic Epilepsy
- In the treatment of primary generalized tonic-clonic seizures in adults and children from 12 years of age with Idiopathic Generalized Epilepsy.

Route of administration: Oral

Recommended Dose

Oral solution

The oral solution may be diluted in a glass of water and may be taken with or without food. The daily dose is administered in two equally divided doses.

Adults

- Monotherapy

Adults and adolescents from 16 years of age

The recommended starting dose is 250 mg twice daily which should be increased to an initial therapeutic dose of 500 mg twice daily after two weeks. The dose can be further increased by 250 mg twice daily every two weeks depending upon the clinical response. The maximum dose is 1500 mg twice daily.

- Add-on therapy

Adults (18 years) and adolescents (12 to 17 years) weighing 50 kg or more

The initial therapeutic dose is 500 mg twice daily. This dose can be started on the first day of treatment.

Depending upon the clinical response and tolerability, the daily dose can be increased up to 1500 mg twice daily. Dose changes can be made in 500 mg twice daily increases or decreases every two to four weeks.

Children

The physician should prescribe the most appropriate pharmaceutical form, presentation and strength according to age, weight and dose.

The tablet formulation is not adapted for use in children under the age of 6 years. Levetiracetam oral solution is the preferred formulation for use in this population. In addition, the available dose strengths of the tablets are not appropriate for initial treatment in children weighing less than 25 kg, for patients unable to swallow tablets or for the administration of doses below 250 mg. In all of the above cases Levetiracetam oral solution should be used.

Monotherapy

The safety and efficacy of levetiracetam in children and adolescents below 16 years as monotherapy treatment have not been established.

There are no data available.

Add-on therapy for children (4 to 11 years) and adolescents (12 to 17 years) weighing less than 50 kg

Levetiracetam oral solution is the preferred formulation for use in children under the age of 6 years.

The initial therapeutic dose is 10 mg/kg twice daily.

Depending upon the clinical response and tolerability, the dose can be increased up to 30 mg/kg twice daily. Dose changes should not exceed increases or decreases of 10 mg/kg twice daily every two weeks.

The lowest effective dose should be used.

Dose in children 50 kg or greater is the same as in adults.

Dose recommendations for children and adolescents:

Weight	**Starting dose** **10 mg/kg twice daily**	**Maximum dose** **30 mg/kg twice daily**
10 kg [1]	100 mg (1 ml) twice daily	300 mg (3 ml) twice daily
15 kg [1]	150 mg (1.5 ml) twice daily	450 mg (4.5 ml) twice daily
20 kg [1]	200 mg (2 ml) twice daily	600 mg (6 ml) twice daily
25 kg	250 mg twice daily	750 mg twice daily
From 50 kg [2])	500 mg twice daily	1500 mg twice daily

[1] Children 25 kg or less should preferably start the treatment with levetiracetam 100 mg/ml oral solution

[2] Dose in children and adolescents 50 kg or more is the same as in adults

Elderly

Adjustment of the dose is recommended in elderly patients with compromised renal function.

Renal impairment

The daily dose must be individualised according to renal function *(see Section Warnings and Precautions).*

For adult patients, refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient"s creatinine clearance (CLcr) in ml/min is needed. The CLcr in ml/min may be estimated from serum creatinine (mg/dl) determination, for adults and adolescents weighting 50 kg or more, using the following formula:

$$\text{CLcr (ml/min)} = \frac{[140\text{- age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \ (\times\ 0.85 \text{ for women})$$

Then CLcr is adjusted for body surface area (BSA) as follows:

$$\text{CLcr (ml/min/1.73 m}^2) = \frac{\text{CLcr (ml/min)}}{\text{BSA subject (m}^2)} \times 1.73$$

<u>Dosing adjustment for adult and adolescent patients weighing more than 50kg with impaired renal function</u>

Group	**Creatinine clearance (ml/min/1.73m^2)**	**Dosage and frequency**
Normal	> 80	500 to 1500 mg twice daily
Mild	50-79	500 to 1000 mg twice daily
Moderate	30-49	250 to 750 mg twice daily
Severe	< 30	250 to 500 mg twice daily

End-stage renal disease patients undergoing dialysis [(1)].	-	500 to 1000 mg once daily [(2)]

[(1)] A 750 mg loading dose is recommended on the first day of treatment with Levetiracetam.

[(2)] Following dialysis, a 250 to 500 mg supplemental dose is recommended.

For children with renal impairment, Levetiracetam dose needs to be adjusted based on the renal function as Levetiracetam clearance is related to renal function.

The CLcr in ml/min/1.73 m^2 may be estimated from serum creatinine (mg/dl) determination, for young adolescents and children using the following formula (Schwartz formula):

$$CLcr\ (ml/min/1.73\ m^2) = \frac{\text{Height (cm)} \times ks}{\text{Serum Creatinine (mg/dl)}}$$

ks= 0.55 in Children to less than 13 years and in adolescent female; ks= 0.7 in adolescent male

Dosing adjustment for children and adolescents patients weighing less than 50 kg with impaired renal function

Group	**Creatinine clearance (ml/min /1.73m²)**	**Children and adolescents weighing less than 50 kg**
Normal	> 80	10 to 30 mg/kg (0.10 to 0.30 ml/kg) twice daily
Mild	50-79	10 to 20 mg/kg (0.10 to 0.20 ml/kg) twice daily
Moderate	30-49	5 to 15 mg/kg (0.05 to 0.15 ml/kg) twice daily
Severe	< 30	5 to 10 mg/kg (0.05 to 0.10 ml/kg) twice daily

End-stage renal disease patients undergoing dialysis	-	10 to 20 mg/kg (0.10 to 0.20 ml/kg)once daily [(2)] [(3)]

[(1)] Levetiracetam oral solution should be used for doses under 250 mg and for patients unable to swallow tablets.

[(2)] A 15 mg/kg (0.15 ml/kg) loading dose is recommended on the first day of treatment with levetiracetam.

[(3)] Following dialysis, a 5 to 10 mg/kg (0.05 to 0.10 ml/kg) supplemental dose is recommended.

Hepatic impairment
No dose adjustment is needed in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the renal insufficiency. Therefore a 50% reduction of the daily maintenance dose is recommended when the creatinine clearance is < 60 ml/min/1.73m^2.

Contraindication

Hypersensitivity to Levetiracetam or other pyrrolidone derivatives or any of the excipients.

Warning and Precautions

“Potential for an increase in risk of suicidal thoughts or behaviours”.

In accordance with current clinical practice, if LEVETIRACETAM has to be discontinued it is recommended to withdraw it gradually (e.g. in adults: 500 mg twice daily decrements every two to four weeks; in children: dose decrease should not exceed decrements of 10 mg/kg twice daily every two weeks).

The administration of LEVETIRACETAM to patients with renal impairment may require dose adaptation. In patients with severely impaired hepatic function, assessment of renal function is recommended before dose selection.

Suicide, suicide attempt and suicidal ideation have been reported in patients treated with levetiracetam. Patients should be advised to immediately report any symptoms of depression and/or suicidal ideation to their prescribing physician.

Acute kidney injury

The use of levetiracetam has been rarely associated with acute kidney injury, with a time to onset ranging from few days to several months.

Interaction with Other Medicaments

Data indicate that LEVETIRACETAM did not influence the serum concentrations of existing antiepileptic medicinal products (phenytoin, carbamazepine, valproic acid, phenobarbital, lamotrigine, gabapentin and primidone) and that these antiepileptic medicinal products did not influence the pharmacokinetics of LEVETIRACETAM.

The clearance of levetiracetam was 22% higher in children taking enzyme-inducing AEDs compared to children who did not take enzyme-inducing AEDs. Dose adjustment is not recommended. Levetiracetam had no effect on plasma concentrations of carbamazepine, valproate, topiramate, or lamotrigine.

Probenecid (500 mg four times daily), a renal tubular secretion blocking agent, has been shown to inhibit the renal clearance of the primary metabolite but not of levetiracetam. Nevertheless, the concentration of this metabolite remains low. It is expected that other drugs excreted by active tubular secretion could also reduce the renal clearance of the metabolite. The effect of LEVETIRACETAM on probenecid was not studied and the effect of LEVETIRACETAM on other actively secreted drugs, e.g. NSAIDs, sulphonamides and methotrexate is unknown.

LEVETIRACETAM 1000 mg daily did not influence the pharmacokinetics of oral contraceptives (ethinylestradiol and levonorgestrel); endocrine parameters (luteinizing hormone and progesterone) were not modified. LEVETIRACETAM 2000 mg daily did not influence the pharmacokinetics of digoxin and warfarin; prothrombin times were not modified. Co-administration with digoxin, oral contraceptives and warfarin did not influence the pharmacokinetics of LEVETIRACETAM.

No data on the influence of antacids on the absorption of LEVETIRACETAM are available. The extent of absorption of LEVETIRACETAM was not altered by food, but the rate of absorption was slightly reduced.

No data on the interaction of LEVETIRACETAM with alcohol are available.

Pregnancy and Lactation

There are no adequate data from the use of LEVETIRACETAM in pregnant women. Studies in animals have shown reproductive toxicity. The potential risk for human is unknown. LEVETIRACETAM should not be used during pregnancy unless clearly necessary.

As with other antiepileptic drugs, physiological changes during pregnancy may affect LEVETIRACETAM concentration. There have been reports of decreased LEVETIRACETAM concentration during pregnancy. Discontinuation of antiepileptic treatments may result in exacerbation of the disease which could be harmful to the mother and the foetus.

LEVETIRACETAM is excreted in human breast milk. Therefore, breast-feeding is not recommended.

Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Due to possible different individual sensitivity, some patients might experience somnolence or other central nervous system symptoms, at the beginning of treatment or following a dose increase. Therefore, caution is recommended in those patients when performing skilled tasks, e.g. driving vehicles or operating machinery.

Side Effects

The most frequently reported adverse reactions were nasopharyngitis, somnolence, headache, fatigue and dizziness. The safety profile of Levetiracetam is generally similar across age groups (adult and paediatric patients) and across the approved epilepsy indications.

Adverse reactions:

<u>Infections and infestations</u>

Very common: nasopharyngitis

Rare: infection

<u>Blood and lymphatic system disorders</u>

Uncommon: thrombocytopenia, leukopenia

Rare: neutropenia, pancytopenia, agranulocytosis

<u>Immune system disorders</u>

Rare: drug reaction with eosinophilia and systemic symptoms (DRESS)

<u>Metabolism and nutrition disorders</u>

Common: anorexia

Uncommon: weight decreased, weight increase

Rare: hyponatraemia

Psychiatric disorders

Common: depression, hostility/aggression, anxiety, insomnia, nervousness/irritability

Uncommon: suicide attempt, suicidal ideation, psychotic disorder, abnormal behavior, hallucination, anger, confusional state, panic attack, affect lability/mood swings, agitation

Rare: completed suicide, personality disorder, thinking abnormal

Nervous system disorders

Very common: somnolence, headache

Common: convulsion, balance disorder, dizziness, lethargy, tremor

Uncommon: amnesia, memory impairment, coordination abnormal/ataxia, paraesthesia, disturbance in attention

Rare: choreoathetosis, dyskinesia, hyperkinesia

Eye disorders

Uncommon: diplopia, vision blurred

Ear and labyrinth disorders

Common: vertigo

Respiratory, thoracic and mediastinal disorders

Common: cough

Gastrointestinal disorders

Common: abdominal pain, diarrhoea, dyspepsia, nausea, vomiting

Rare: pancreatitis

Hepatobiliary disorders

Uncommon: liver function test abnormal

Rare: hepatic failure, hepatitis

Skin and subcutaneous tissue disorders

Common: rash

Uncommon: alopecia, eczema, pruritus

Rare: toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme

Musculoskeletal and connective tissue disorders

Uncommon: muscular weakness, myalgia

Rare: Rhabdomyolysis and blood creatine phosphokinase increased

Prevalence is significantly higher in Japanese patients when compared to non-Japanese patients.

Cases of encephalopathy have been rarely observed after levetiracetam administration. These undesirable effects generally occurred at the beginning of the treatment (few days to a few months) and were reversible after treatment discontinuation.

General disorders and administration site conditions

Common: asthenia/fatigue

Injury, poisoning and procedural complications

Uncommon: injury

Renal & urinary disorders

Rare: Acute kidney injury

Description of selected adverse reactions

The risk of anorexia is higher when topiramate is coadministered with Levetiracetam. In several cases of alopecia, recovery was observed when Levetiracetam was discontinued. Bone marrow suppression was identified in some of the cases of pancytopenia.

Symptoms and Treatment of Overdose

Symptoms: Somnolence, agitation, aggression, depressed level of consciousness, respiratory depression and coma were observed with LEVETIRACETAM overdoses.

Management of overdose: After an acute overdose, the stomach may be emptied by gastric lavage or by induction of emesis. There is no specific antidote for levetiracetam. Treatment of an overdose will be symptomatic and may include haemodialysis. The dialyser extraction efficiency is 60% for levetiracetam and 74% for the primary metabolite.

Storage

Store below 30ºC.

Shelf Life

3 years

Presentation

LETRAM OS (LEVETIRACETAM ORAL SOLUTION 100 MG/ML) 150ml pack size filled in 200 ml glass bottle/28 mm neck amber TYPE III-DMF, capped with child resistant with expanded PE wad, 28 mm.

LETRAM OS (LEVETIRACETAM ORAL SOLUTION 100 MG/ML) 300ml pack size filled in 300 ml glass bottle/28 mm neck amber TYPE III-DMF, capped with child resistant with expanded PE wad, 28 mm.

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